6.3 Pascal's Triangle – Worksheet #2

MCR3U

SOLUTIONS

1) Expand each expression using Pascal's triangle

a) $(x+4)^3$

$= 1x^3 + 3x^2(4) + 3x(4^2) + 1(4^3)$

$= x^3 + 12x^2 + 48x + 64$

b) $(1-2x)^4$

$= 1(1^4) + 4(1^3)(-2x) + 6(1^2)(-2x)^2 + 4(1)(-2x)^3 + 1(-2x)^4$

$= 1 - 8x + 24x^2 - 32x^3 + 16x^4$

c) $(3x+y)^2$

$= 1(3x)^2 + 2(3x)(y) + 1(y)^2$

$= 9x^2 + 6xy + y^2$

d) $(x-5)^5$

$= 1(x)^5 + 5(x)^4(-5) + 10(x)^3(-5)^2 + 10(x)^2(-5)^3 + 5(x)(-5)^4 + 1(-5)^5$

$= x^5 - 25x^4 + 250x^3 - 1250x^2 + 3125x - 3125$

e) $(3+2n)^6$

$= 1(3)^6 + 6(3)^5(2n) + 15(3)^4(2n)^2 + 20(3)^3(2n)^3$
$+ 15(3)^2(2n)^4 + 6(3)(2n)^5 + 1(2n)^6$

$= 729 + 2916n + 4860n^2 + 4320n^3$
$+ 2160n^4 + 576n^5 + 64n^6$

f) $(x-7)^{11}$

skip.

g) $(2x + 5)^7$

$= 1(2x)^7 + 7(2x)^6(5) + 21(2x)^5(5)^2 + 35(2x)^4(5)^3 + 35(2x)^3(5)^4 + 21(2x)^2(5)^5 + 7(2x)(5)^6 + 1(5)^7$

$= 128x^7 + 2240x^6 + 16800x^5 + 70000x^4 + 175000x^3 + 262500x^2 + 218750x + 78125$

Answers

1. a) $x^3 + 12x^2 + 48x + 64$
b) $1 - 8x + 24x^2 - 32x^3 + 16x^4$
c) $9x^2 + 6xy + y^2$
d) $x^5 - 25x^4 + 250x^3 - 1250x^2 + 3125x - 3125$

e) $729 + 2916n + 4860n^2 + 4320n^3 + 2160n^4 + 576n^5 + 64n^6$

f) $x^{11} - 77x^{10} + 2695x^9 - 56\,595x^8 + 792\,330x^7 - 7\,764\,834x^6 + 54\,353\,838x^5 - 271\,769\,190x^4 + 951\,192\,165x^3 - 2\,219\,448\,285x^2 + 3\,107\,227\,739x - 1\,977\,326\,743$

g) $128x^7 + 2240x^6 + 16\,800x^5 + 70\,000x^4 + 175\,000x^3 + 262\,500x^2 + 218\,750x + 78\,125$